

CHANDRIMA BHATTACHARYYA

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SUMMARY

Pre-final year Electronic & Computer Engineering student (Integrated Master, Trinity College Dublin) with hands-on experience in FPGA development, RTL design, embedded systems, and hardware-software integration. Experienced in implementing synthesizable FSM architectures, timing constraints, and real-time signal processing on Xilinx BASYS3 (Artix-7) using Verilog/VHDL and Vivado. Strong foundation in Digital Systems Design, Embedded C/C++, Python automation, MySQL-backed applications, and sensor-driven hardware systems. Passionate about semiconductor, FPGA, and ASIC-oriented engineering roles.

EDUCATION

Trinity College Dublin <i>Integrated Master in Electronic & Computer Engineering (Transferred)</i>	Sep 2025 – Sep 2028 Year Grade: 2.1
Thapar Institute of Engineering & Technology, Patiala <i>B.E. Electronic & Computer Engineering (2+2 Transfer)</i>	Aug 2023 – Aug 2025 Cumulative GPA: 8.35/10.0

KEY MODULES

Trinity College Dublin – Year 3	
Digital Systems Design – I	Circuits & Systems Design – I
Sensors – II.1	Signals & Systems – II.1
Data Structures & Algorithms – II.1	Probability & Statistics – II.1
Electronic Circuits – II.2	Engineering Electromagnetics – II.2
Thapar Institute of Engineering & Technology – Year 1–2	
OOPs – A	Embedded Systems – A
Numerical Linear Algebra – A	Signals & Systems – A-
Analog Circuits – A-	C Programming – A-
Digital Electronics – B	Computer & Communication Networks – B

TECHNICAL SKILLS

HDL / FPGA: Verilog, VHDL, Vivado (Xilinx), FSM design, timing constraints, AXI4-Stream / AXI-Lite, Vivado ILA (Integrated Logic Analyzer); BASYS3 (Artix-7) target board

Embedded: Embedded C/C++ (ATmega328P, ARM), ARM Assembly, PID control, RTOS concepts, Keil µVision, Arduino IDE

EDA / PCB: EagleCAD, LTSpice, AutoCAD, SolidWorks; dual-layer PCB fabrication & chemical etching

Programming: Python, Embedded C/C++, MATLAB, SQL/MySQL, Python Tkinter

Data / Productivity: Excel (Pivot Tables, Reporting), PowerPoint, Word, pandas, NumPy, matplotlib

Protocols: PCIe (basics), Ethernet (MAC/PHY fundamentals), UART, SPI, I2C, wireless RF modules (NRF/similar), MQTT

Tools: Git/GitHub, Linux CLI, Vivado Simulator

Lab Instrumentation: Oscilloscopes, Logic Analyzers, Multimeters, Voltmeters, Spectrum Analyzers

Languages: English (fluent), Bengali (fluent), Hindi (fluent), German (A2)

PROJECTS

MAC: Multi-Sensor Interactive Gaming Console <i>Group Project</i>	Apr 2026
- Designed and built an Arduino Uno-based multi-game embedded platform integrating an HC-SR04 ultrasonic sensor, a glove-mounted film pressure sensor (voltage-divider conditioned, 0.1 µF RC filter), a rotary potentiometer, and a passive speaker - all driven from a single ATmega328P microcontroller. - Implemented a menu-driven finite state machine in Embedded C++ spanning calibration, menu navigation, and three independent game states (Flappy Bird, Tone Matching, Skiing); calibration averages sensor readings over fixed intervals and validates flex thresholds before admitting users. - Engineered a pseudo-3D perspective rendering pipeline for the Skiing game on a 128×64 SSD1306 OLED (4-wire SPI): per-frame perspective projection ($s_x = CX + (wx/Z) \cdot F$), depth-sorted gate rendering, procedural random-walk track curvature, dithered snow texture, three-layer parallax environment, and a 160-byte PROGMEM mountain bitmap - achieving ~50 fps within a 1,024-byte SRAM framebuffer. - Stored per-user high scores in EEPROM for persistence across power cycles; managed shared 2 KB SRAM across display buffer, game state, and sensor data with zero dynamic allocation.	

- Transitioning from Arduino Uno to a higher-performance MCU architecture to support concurrent multi-sensor data fusion and audio synthesis with sub-10 ms sensor-to-output latency.

FPGA & VLSI Development on BASYS3 - FSM & Real-Time Signal Processing

Apr 2026

Self Project

- Implementing synthesisable multi-state FSMs in Verilog/VHDL targeting the Xilinx Artix-7 (BASYS3) using Vivado; applying timing constraints and static timing analysis to eliminate latency bottlenecks in real-time signal and audio processing paths.
- Leveraging deterministic hardware execution and clock-domain management to replace interrupt-driven MCU limitations; progressing toward hardware-accelerated multi-sensor data fusion.

Autonomous Buggy - PID Line Following & Self-Parking

Jun 2025

Group Project

- Programmed ATmega328P in Embedded C++ to execute closed-loop PID line following using calibrated IR sensor arrays; achieved stable tracking at 0.5 m/s with corner overshoot under 3 cm.
- Implemented a self-parking state machine with encoder feedback; buggy completed a 5-point turn manoeuvre within a 40 cm corridor.

IoT Remote Monitoring System - End-to-End Data Pipeline

Jun 2025

Group Project

- Deployed an IoT system integrating environmental sensors over MQTT/HTTP; built a data-logging backend and live visualisation dashboard with under 2 s telemetry update latency.

Mangonel Arm Position Sensing & Control

Oct 2024

Group Project

- Integrated rotary position sensors with embedded control logic for real-time angular kinematic feedback; validated accuracy to within $\pm 2^\circ$ across a 180° sweep.

DC Power Supply & Analog Circuitry

Oct 2024

Self Project

- Designed and assembled a stabilised 5 V DC supply from 230 V AC mains: full-wave rectifier, LC filter, and Zener-diode regulation; measured output ripple < 50 mV under 500 mA load.

Custom Dual-Layer PCB Design & Fabrication

Oct 2023

Self Project

- Designed schematic and dual-layer PCB layout in EagleCAD; fabricated and chemically etched boards in-lab, integrating wireless RF modules for command/telemetry links.

Digital Image Processing - Geometric Shape Detection (Python)

Feb 2024

Self Project

- Built a Python/OpenCV pipeline for real-time shape classification using Canny edge detection and contour analysis; achieved $> 92\%$ accuracy on a 200-image test set.

Genius© - A Notemaking App (Python GUI, Tkinter, MySQL)

Feb 2023

Self Project

- Built a Python/MySQL database backed application using Python, Tkinter for an interactive front-end for real-time shape task management using Tkinter as the GUI.

ACTIVITIES & LEADERSHIP

SPICMACAY, TIET Chapter

Sep 2023 – Jun 2025

Society for the Promotion of Indian Classical Music and Culture Amongst Youth

Editorial Board Member

Jan 2025 – Jun 2025

- Led editorial strategy, marketing campaigns, and event budgeting, growing chapter reach to 2,000+ followers.

Member

Sep 2023 – Jan 2025

- Contributed to graphic design, media promotion, and fundraising initiatives for chapter events.

Robotics Club

2015 – 2018

Presidium School

- Active learning experience in robotics fundamentals, teamwork, and hands-on building.

Choir

2014 – 2020

Presidium School

- Competed in several events across the National Capital Territory.